

BCSE 1593: DATA SCIENCE AND VISUALIZATION LAB

Objective: The objective of the course is to enable students to understand and use python and other tools for Data Science. Students learn how to analyze data.

Credits: 1

L–T–P–J: 0-0-2-0

Module No.	Content	Teaching Hours
I	Exploring Kaggle & GitHub: Create accounts, download a dataset from Kaggle, and upload analysis to GitHub repository. Numpy: Use NumPy ndarray to analyze dataset (indexing, slicing, reshaping, stats). Pandas: Work with Series/DataFrame to analyze dataset (filtering, grouping, summarizing). Data Loading & Storage: CSV, Excel, JSON, HTML, Load datasets from CSV/Excel/JSON and parse .html using Pandas.	10
II	Web API Interaction: Fetch JSON data from a public API and analyze results with Pandas. Data Cleaning: Handle missing, duplicate, and inconsistent values in dataset. Data Wrangling: Merge, join, group, and reshape dataset for structured analysis. Data Visualization: Plot line, bar, scatter, and histogram charts with labels and legends. Time Series Analysis: Work with datetime-indexed sales data, resampling, and trend visualization.	10

Text Book:

- McKinney, W. (2017). Python for Data Analysis: Data Wrangling with Pandas, NumPy and IPython. 2nd edition. O'Reilly Media
- O'Neil, C., & Schutt, R. (2013). Doing Data Science: Straight Talk from the Frontline, O'Reilly Media

Reference Books:

- VanderPlas, J. (2016). Python data science handbook: Essential tools for working with data. O'Reilly Media, Inc.

Outcome:

After completion of course, student will be able to:

- Understand the basics of data analytics
- Learn how to use python for data analytics
- Learn how to sort and filter data in python
- Learn how to use lookup functions in python
- Learn how to use conditional formatting in python
- Learn how to create pivot tables in python